

Aditya Dasika

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Production AI/ML engineer specializing in agentic and generative AI systems. At Phenom People, architected chatbotCXAgent, a 12-node LangGraph orchestration platform handling intent recognition, dialogue state management, tool-routing, and Graph RAG retrieval, shipped to production users. Built the full agentic stack end to end: LangChain/LangGraph orchestration, FAISS-based RAG, OpenAI and Anthropic API integration, and XGrammar-enforced tool-calling reliability (91.9% routing accuracy, zero malformed calls in production). Strong core Python fundamentals (OOP, decorators, typing, data structures) developed across 4+ years of production software engineering, including QLoRA fine-tuning on vLLM (10.98 RPS at 98.6% success) and multi-task NLP with PyTorch. Built with Claude Code.

Core Competencies

- **Generative & Agentic AI:** LangGraph (production orchestration, LangGraph agents), LangChain, RAG and Graph RAG, FAISS vector search, OpenAI & Anthropic APIs, tool/function calling, XGrammar constrained decoding
- **Core Python:** OOP, decorators, typing/dataclasses, iterators/generators, exception handling, design patterns, data structures & algorithms, applied across 4+ years of production systems
- **ML & NLP:** PyTorch, Transformers (BERT/ModernBERT), multi-task learning, NLU/NER/intent recognition, NumPy, Pandas, classical forecasting models
- **LLM Fine-Tuning & Serving:** QLoRA, SFT, Unsloth, HuggingFace PEFT, NF4 quantization, vLLM, Ollama
- **Eval & Observability:** pass@k, LLM-as-judge rubrics, LiteLLM, Langfuse, production MLOps
- **Backend & Cloud:** FastAPI, Node.js, Docker, AWS (EC2/S3/SageMaker), PostgreSQL, SQL, CI/CD; cloud ML pipeline patterns (training, serving, cost monitoring) built on AWS, transferable to GCP-equivalent services

Professional Experience

- **Phenom People** **Hyderabad, Telangana**
AI/LLM Engineer II *Apr 2025–Present*
 - Architected **chatbotCXAgent**, a production **agentic** platform on **LangGraph** (12 orchestrated nodes) handling intent recognition, dialogue state management, tool-routing, and **Graph RAG** retrieval across enterprise workflows, shipped to real users in production
 - Enforced tool-calling reliability with XGrammar constrained decoding and schema-validated JSON outputs, raising tool-routing accuracy to **91.9%** and eliminating malformed tool calls in production
 - Fine-tuned and served Qwen models via QLoRA (Unsloth, NF4) on vLLM / AWS EC2 g5.xlarge (A10G), sustaining **10.98 RPS at 98.6% success** under load and roughly halving inference cost vs. hosted GPT-4.1
 - Built a custom eval framework (pass@k, LLM-judge rubric) gating every pipeline change across all 12 nodes before release; instrumented LiteLLM + Langfuse tracing for per-node cost and latency observability
 - Fine-tuned a multi-task **PyTorch** ModernBERT model (intent classification + NER) improving NLU accuracy from **87% to 97%** via differential learning rates, label smoothing, and uncertainty-weighted multi-task loss
- **Blueleaves Farms** **Hyderabad, Telangana**
Full Stack Engineer *May 2022–Feb 2025*
 - Built data-driven forecasting and analytics, including demand estimation, sales-trend, and inventory-planning models on transaction and stock-movement data, improving procurement and replenishment decisions
 - Owned end-to-end delivery of production systems across POS, CRM, inventory, procurement, and e-commerce, serving **5+ locations and 10,000+ monthly transactions** while cutting manual operational errors by ~90%
 - Designed scalable backend workflows on Firebase Cloud Functions, Node.js, and REST APIs, including invoicing, inventory synchronization, order lifecycle, and role-based access control, integrating multiple internal systems end to end

Projects

- **Neo4j, Ollama, PyMuPDF, Python, Docker**
IndexNotes — AI Knowledge Graph for Scientific Papers
 - Built an automated **RAG** pipeline ingesting scientific papers (PDF to local-LLM extraction to structured JSON to Neo4j), creating 40+ concepts and 17+ typed relationships across 3 research papers, designed for **Graph RAG** queries
- **PyTorch, Autograd, L-BFGS, Latin Hypercube Sampling**
Physics-Informed Neural Networks (PINNs)
 - Implemented PINNs from scratch in **PyTorch** solving 3 PDEs: Heat equation ($\sim 3\%$ error), Burgers' equation (matching Raissi et al.), and Schrödinger equation (complex-valued, final loss 0.000083)

Education

- **Andhra Loyola Institute of Engineering and Technology**
Bachelor of Technology in Mechanical Engineering

Vijayawada, Andhra Pradesh
Aug 2018–May 2022

Languages

- English (fluent), Telugu (native)

References

- Available upon request.